PLAYLISTGENERATOR

Semantic Music Retrieval from Natural Language and Visual Prompts

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CONTENTS

1 INTRODUCTION

Music streaming platforms host hundreds of millions of tracks, yet their recommendation engines still rely largely on collaborative filtering and rigid keyword tags. When a listener wants "*chill jazz for relaxing*" or "*upbeat reggae with rhythmic percussion*", a keyword search often fails to match the intended mood, because natural language encodes semantic nuance that traditional metadata cannot capture.

This project, **PlaylistGenerator**, addresses this gap by building a semantic music retrieval system that maps both audio and free-text queries into a shared embedding space. Given any natural-language prompt, the system returns a ranked playlist of the most *semantically* similar tracks in the corpus. An additional research direction extends this to *visual* prompts: a user-provided image can be projected into the same embedding space, enabling image-to-playlist queries.

The system is built on three components:

1. **Audio corpus indexing** — pre-compute CLAP audio embeddings for every track in the FMA Small dataset and store them in a compressed index.
2. **Text-to-playlist retrieval** — encode the user's text prompt with the CLAP text encoder and run a cosine k-nearest-neighbor search against the index.
3. **Domain adaptation** — fine-tune the CLAP text encoder with LoRA on the MusicCaps dataset to better align music-specific captions with audio.

A fourth, exploratory component trains a small MLP bridge from CLIP image features into the CLAP text space, opening the path toward image-conditioned playlist generation.

2 BACKGROUND

2.1 Contrastive Language–Audio Pre-training (CLAP)

CLAP [**?**] is a dual-encoder model that maps audio clips and text descriptions into a shared 512-dimensional embedding space via contrastive learning. The audio branch uses an **HTSAT-base**

hierarchical Swin transformer, while the text branch is **RoBERTa**. During pre-training, matched audio–text pairs are pulled together while mismatched pairs are pushed apart using the **symmetric InfoNCE loss**:

$$\mathcal{L} = \frac{1}{2}\Big(\mathcal{L}_{\text{audio}\to\text{text}} + \mathcal{L}_{\text{text}\to\text{audio}}\Big), \quad \mathcal{L}_{a\to t} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{s\,\mathbf{a}_i^\top \mathbf{t}_i}}{\sum_{j=1}^{N} e^{s\,\mathbf{a}_i^\top \mathbf{t}_j}} \tag{1}$$

where $s = \exp(\log_\text{scale})$ is a learned temperature and N is the batch size. The checkpoint used throughout this project is `music_audioset_epoch_15_esc_90.14.pt`, pre-trained on music and AudioSet data, achieving 90.14% accuracy on ESC-50.

2.2 CLIP for Vision

CLIP [**?**] follows the same contrastive paradigm for image–text pairs. We use the **ViT-L/14** variant (`openai/clip-vit-large-patch14`), which produces 1024-dimensional image embeddings. By training a bridge network from CLIP's image space to CLAP's text space, we can query the audio index with images instead of text.

2.3 Parameter-Efficient Fine-tuning with LoRA

LoRA [**?**] inserts low-rank adapters into the frozen weight matrices of a pre-trained model. For a weight matrix $W \in \mathbb{R}^{d\times k}$, the adapted forward pass becomes:

$$h = Wx + \Delta W\,x = Wx + BA\,x, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d,k). \tag{2}$$

Only A and B are trained; the original weights W remain frozen. With $r = 16$ and $\alpha = 32$, injecting LoRA into the `query` and `value` projections of RoBERTa yields only **1,246,209 trainable parameters**, out of 200,154,457 total (0.62%).

3 DATASETS

3.1 FMA Small

The **Free Music Archive (FMA) Small** subset [**?**] contains **8,000 tracks** across 8 balanced genres (Hip-Hop, Pop, Folk, Experimental, Rock, International, Electronic, Instrumental), each 30-second clip at 44.1 kHz. The archive weighs approximately 7.2 GiB. Out of 8,000 files, 6 are corrupt MP3s (silently skipped during embedding), yielding **7,994 indexed tracks**. This corpus constitutes the retrieval database.

3.2 MusicCaps

MusicCaps [**?**] is a music captioning benchmark released by Google and hosted on Hugging Face (`google/MusicCaps`). It provides 5,521 10-second YouTube clips paired with expert free-text captions describing instrumentation, tempo, mood, and genre. After downloading with `yt-dlp`, 5,298 clips were successfully retrieved (223 permanently unavailable). Pre-computing audio embeddings from the frozen CLAP audio encoder and pairing them with captions produced **4,357 usable samples**, split 85 / 15 into **3,704 training** and **653 test** samples (seed 42). This split is used for LoRA fine-tuning.

3.3 MS COCO Captions

The **MS COCO Captions** dataset provides image–caption pairs for the image-conditioned bridge experiment. We sample the first **20,000 training examples** (shuffled, seed 42) from the Hugging Face `HuggingFaceM4/COCO` dataset. Each sample provides one image (encoded by the frozen CLIP ViT-L/14) and one caption (encoded by the frozen CLAP text encoder), yielding paired $(v_{\text{CLIP}}, e_{\text{CLAP}})$ tuples for bridge MLP training.

4
SYSTEM ARCHITECTURE AND METHODS

4.1 Overview

Figure **??** gives a high-level view of the three inference paths the system supports.

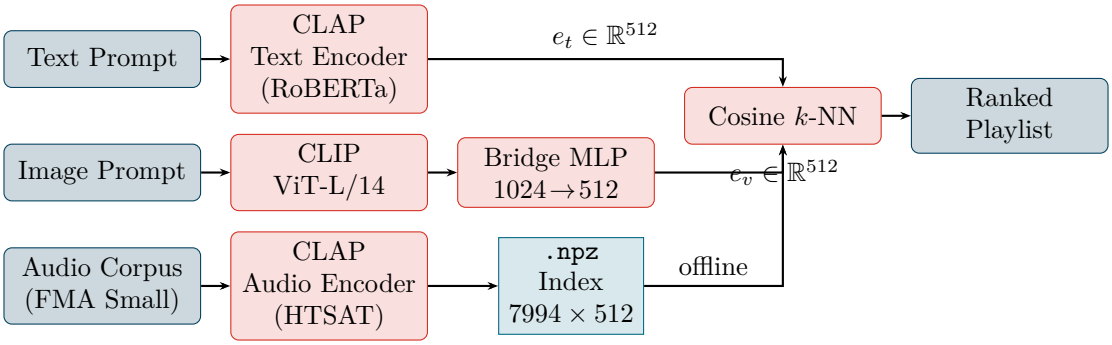

Figure 1: System overview. Text and image prompts are encoded into the 512-dim CLAP space; the audio corpus is indexed offline; a cosine k-NN retrieves the top-k most similar tracks.

4.2 Audio Corpus Indexing

All 8,000 MP3 files in the FMA Small dataset are processed with the CLAP audio encoder (`compute_embeddings.ipy` Files are loaded at 22050 Hz, passed through the `CLAP_Module.get_audio_embedding_from_filelist` method, then L2-normalised. Six corrupted files raise a `NoBackendError` and are silently skipped. The resulting **7,994 vectors of dimension 512** are stored in a compressed NumPy archive (`data/embeddings.npz`), along with their file-path keys, constituting the *audio index*.

4.3 Text-to-Playlist Retrieval

At inference time (`utils.get_playlist`), the user's natural language query is encoded by the CLAP text encoder and L2-normalised to produce a query vector $e_q \in \mathbb{R}^{512}$. Cosine similarity with all corpus embeddings reduces to a matrix–vector product:

$$\mathrm{sim}(i) = e_q^\top e_i, \quad e_i \text{ already L2-normalised.} \tag{3}$$

The top-k indices are extracted in $\mathcal{O}(N)$ via `np.argpartition`, then sorted. The function returns a list of `(score, filepath)` pairs.

4.4 LoRA Fine-tuning on MusicCaps

The pre-trained CLAP model was trained on general audio–text pairs, so its text encoder may underperform on music-specific captions. We fine-tune *only the text encoder* (RoBERTa) using LoRA while keeping the audio encoder **fully frozen**.

Adapter configuration. `LoraConfig(r=16, lora_alpha=32, lora_dropout=0.1, target_modules=["quer` is applied to `model.model.text_branch` via PEFT, yielding 1,246,209 trainable parameters (0.62% of 200M).

Training procedure. Audio embeddings are pre-computed once from the frozen audio encoder and cached as `data/musiccaps_embeddings.npz`. Each training step loads a batch of *(cached audio embedding, caption)* pairs; the text encoder encodes the captions to produce e_t, and the symmetric InfoNCE loss (Eq. **??**) is computed between the batch of audio embeddings e_a and text embeddings e_t. Hyperparameters are summarised in Table **??**.

Table 1: LoRA fine-tuning hyperparameters.

Hyperparameter	**Value**
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
LR scheduler	Cosine annealing
Batch size	16
Epochs	5
Gradient clipping	norm ≤ 1.0
LoRA rank r	16
LoRA α	32
LoRA dropout	0.1
Target modules	`query`, `value`

4.5 CLIP→CLAP Bridge MLP

To support image-conditioned queries, a small MLP (`BridgeMLP`) is trained to project 1024-dimensional CLIP image features into the 512-dimensional CLAP text space. Both the CLIP and CLAP encoders are **fully frozen**; only the bridge is trained. The architecture is:

$$
\begin{aligned}
h_1 &= \mathrm{Dropout}_{0.2}(\mathrm{LayerNorm}(\mathrm{ReLU}(W_1 x)))\,, \quad W_1 \in \mathbb{R}^{1024\times 1024} \\
h_2 &= \mathrm{Dropout}_{0.2}(\mathrm{LayerNorm}(\mathrm{ReLU}(W_2 h_1)))\,, \quad W_2 \in \mathbb{R}^{1024\times 1024} \\
\hat{e} &= W_3 h_2, \quad W_3 \in \mathbb{R}^{1024\times 512}.
\end{aligned}
$$

Training uses a one-directional InfoNCE loss at temperature $\tau = 0.07$ on 20,000 MS COCO caption pairs, with AdamW ($lr = 2 \times 10^{-4}$, $\lambda = 10^{-2}$), batch size 64, 5 epochs. The trained bridge is saved to `artifacts/bridge_mlp_clip_to_clap.pt`.

5 EXPERIMENTS AND RESULTS

5.1 LoRA Fine-tuning Results

Table **??** reports the symmetric InfoNCE loss on the training and test splits of MusicCaps at the end of each epoch.

Table 2: Symmetric InfoNCE loss per epoch on MusicCaps (batch size 16, 3704/653 samples).

Epoch	Train loss	Val loss
1	0.7132	0.5989
2	0.4678	0.5349
3	0.3770	0.5230
4	0.3296	0.5056
5	0.3086	0.5035

The training loss decreases steadily from 0.71 to 0.31 across five epochs. The validation loss converges to 0.50, indicating that the model generalises well without overfitting despite the small number of trainable parameters. The gap between training and validation loss is consistent with the low model complexity introduced by LoRA.

5.2 Cosine Similarity Before and After Fine-tuning

To directly measure alignment quality, we compute the average cosine similarity between audio embeddings and their corresponding ground-truth text captions on the MusicCaps test split.

Table 3: Average cosine similarity on the MusicCaps test set (653 samples).

Model	Avg. cosine similarity	Relative gain
Pre-trained CLAP (baseline)	0.3161	—
CLAP + LoRA (fine-tuned)	0.3524	+11.5%

Fine-tuning with LoRA raises the average alignment score from 0.3161 to 0.3524, a **+0.0363 absolute** (+11.5% **relative**) improvement. This confirms that domain adaptation of the text encoder to music-specific language significantly improves audio–text correspondence, even when only 0.62% of parameters are updated.

5.3 Qualitative Retrieval Results

Table **??** shows the top-5 retrieved MusicCaps tracks for three representative prompts using the fine-tuned model. The cosine scores range from 0.38 to 0.42, and the retrieved captions are semantically consistent with the query.

Table 4: Top-5 retrieval results from MusicCaps for three prompts (fine-tuned CLAP + LoRA).

Rank	**Caption (truncated)**	**Score**
Prompt: "slow chill hip hop with a mellow bassline"		
1	Hip hop song with consistent low-quality bassline	0.4048
2	Dark moody piano melody with hip hop beat	0.3923
3	Instrumental movie soundtrack theme	0.3897
4	Electronic instrumental at medium tempo	0.3713
5	Instrumental jazz tune with medium-paced bass	0.3684
Prompt: "upbeat reggae with rhythmic percussion"		
1	Acoustic drum playing a reggae groove with rim shot	0.4213
2	Female singer over a cool melody	0.4133
3	Medium-sized band with rhythmic instrumental section	0.3963
4	Electronic instrumental clip	0.3898
5	Reggae song with flat male vocal rapping	0.3844
Prompt: "sad orchestral strings and piano"		
1	Sustained strings melody, low quality recording	0.4070
2	Classical wide mellow strings melody	0.4022
3	Mellow piano melody, low quality recording	0.3986
4	Mellow piano melody, low quality recording	0.3949
5	Sustained string melody, low quality recording	0.3912

The retrieval results exhibit strong semantic coherence: all hip-hop-related captions appear in the top-5 for the first prompt, all reggae-related clips for the second, and all string/piano-based classical or orchestral tracks for the third.

5.4 FMA Small Playlist Demo

For the primary use case, the system is deployed against the FMA Small index (7,994 tracks). Table **??** shows the top-5 results for the query *"Chill jazz for relaxing"* (out of $k = 20$ returned tracks).

Scores above 0.50 indicate high confidence matches. The complete 20-track ranked list is rendered as an in-notebook HTML audio player for direct preview. When the LoRA adapter is loaded

Table 5: Top-5 playlist for *"Chill jazz for relaxing"* from the FMA Small index.

Rank	Track file	Cosine score
1	149142.mp3	0.5401
2	115471.mp3	0.5143
3	074373.mp3	0.5126
4	125621.mp3	0.4839
5	070302.mp3	0.4797

(`USE_LORA=True`), scores shift slightly due to the refined text embedding; the adapter is loaded from `checkpoints/clap_lora/` via `utils.load_clap_model_with_lora`.

6 CONCLUSION AND FUTURE WORK

This project demonstrates that a pre-trained CLAP model can serve as a practical backbone for semantic music retrieval, and that lightweight LoRA fine-tuning on the music-specific MusicCaps benchmark meaningfully improves alignment quality (+11.5% cosine similarity) while updating less than 1% of the model's parameters. The result is a working end-to-end system: given a free-text description, the user receives a ranked playlist drawn from 7,994 FMA tracks within milliseconds.

Two main directions are planned for future work.

Image-to-playlist generation. The trained CLIP→CLAP bridge MLP enables a new interaction modality: a user uploads a photograph (a forest, a party scene, a painting) and the system projects the CLIP image embedding into the CLAP text space, then queries the audio index directly. The bridge is trained but not yet integrated into the end-to-end retrieval pipeline; this integration is the immediate next step.

RLHF alignment. Cosine similarity in the CLAP space is a proxy for human preference, not a direct measure of it. Reinforcement Learning from Human Feedback (RLHF) could be used to further refine the retrieval ranking: users listen to suggested tracks, provide positive/negative signals, and a reward model is trained to score playlists according to human satisfaction. This would close the loop between the embedding-based retrieval and actual listener preferences, making the system more personalised over time.

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